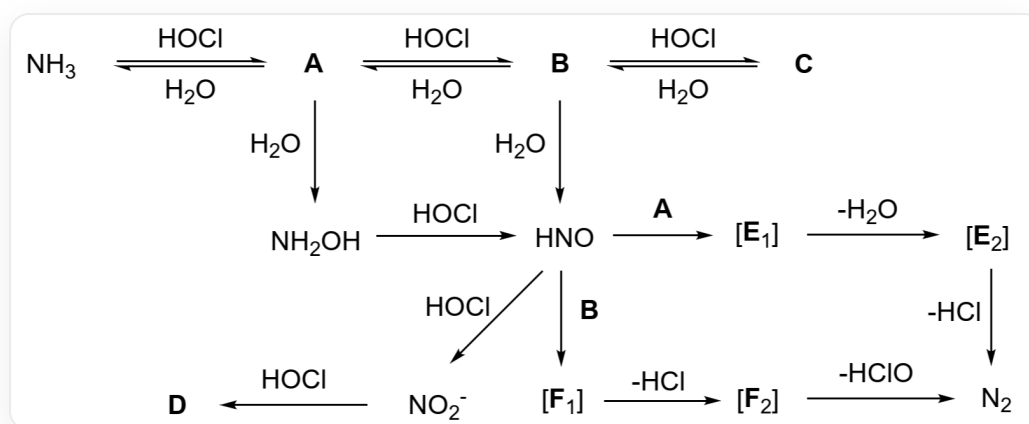


Question

Ammonia and chlorine are two common industrial gases with complex interconversion relationships. For example, chlorine can react with ammonia to produce nitrogen gas, a method often used to remove ammonia from water. The diagram below illustrates the interconversion relationships in the chlorine-ammonia system.



The image is a flowchart of chemical reactions, with the specific process described as follows: Ammonia reversibly forms compound **A** in the presence of hypochlorous acid and water. **A** reversibly forms compound **B** in the presence of hypochlorous acid and water. **B** reversibly forms compound **C** in the presence of hypochlorous acid and water. **A** reacts with water to form hydroxylamine, which reacts with hypochlorous acid to form HNO. **B** also reacts with water to form HNO. HNO reacts with hypochlorous acid to form nitrite ions, and nitrite ions react with hypochlorous acid to form compound **D**. There are two pathways for HNO to form nitrogen gas: Pathway 1 involves HNO reacting with compound **A** to form intermediate **E**₁, which eliminates a water molecule to form intermediate **E**₂, and **E**₂ eliminates a hydrogen chloride molecule to form nitrogen gas. Pathway 2 involves HNO reacting with compound **B** to form intermediate **F**₁, which eliminates a hydrogen chloride molecule to form intermediate **F**₂, and **F**₂ eliminates a hypochlorous acid molecule to form nitrogen gas. The reactions forming compounds **A**, **B**, **C** are reversible, while the other reactions are unidirectional.

Regarding the unknown nitrogen-containing compounds **A**, **B**, **C**, **D** and reaction intermediates **E**₁, **E**₂, **F**₁, **F**₂ in the diagram, the following statements are made:

1. The relative molecular masses of **A**, **B**, **C** increase sequentially.
2. **D** can be used for agricultural purposes.

3. If atoms of different elements are treated as spheres of the same size, the absolute geometric configurations of **A**, **B**, **C** are identical.
4. **E**₁ has more conformational isomers than **E**₂.
5. **F**₁ contains a π bond.
6. **E**₂ has only two configurational isomers.
7. Hydrolysis of **C** can yield **D**.
8. According to the diagram, **A** and **B** can react with HNO to produce nitrogen gas, and **C** can similarly react with HNO via a mechanism analogous to that of **A** and **B** to produce nitrogen gas.

Which of the following options contains all correct statements?

A. 1, 2, 3, 4, 5

B. 1,3,4,5,6

C. 1, 2, 3, 4, 6

D. 2, 3, 4, 6, 7

E. 1, 3, 4, 6, 8

F. 1,4,6,7,8

G. 2, 5, 6, 7, 8

Answer

Correct Answer: **C**

Detailed Explanation

Let us first deduce the identities of these eight unknown substances.

The reactions generating **A**, **B**, **C** are clearly the stepwise chlorination of ammonia by hypochlorous acid,

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Stepwise chlorination of ammonia by hypochlorous acid

where hypochlorous acid acts as the oxidizing agent. Thus, **A** is NH_2Cl , **B** is NHCl_2 , and **C** is NCl_3 . Therefore, Statement 1 is correct.

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Statement 1 is correct

The reaction generating **D** is also straightforward. Nitrite reacts with the strong oxidizing agent hypochlorous acid, and the oxidation product is clearly nitric acid. Since the question specifies that **D** is a compound, it cannot be interpreted as nitrate here,

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D is specified as a compound and cannot be interpreted as nitrate here

so **D** is HNO_3 . Nitric acid can be used to prepare its ammonium salt for nitrogen fertilizer, so Statement 2 is correct.

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Statement 2 is correct

According to VSEPR theory, **A, B, C** are all trigonal pyramidal molecules. If atoms of different elements are treated as spheres of the same size, the absolute geometric configurations of **A, B, C** are identical, so Statement 3 is correct.

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A, B, C are trigonal pyramidal

CHECKPOINT

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Statement 3 is correct

NCl_3 hydrolyzes to form nitrogen gas, hypochlorous acid, and hydrochloric acid, but not nitric acid, so Statement 7 is incorrect.

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Statement 7 is incorrect

The reaction of HNO with compounds **A**, **B** is essentially an addition reaction where the $\text{N}-\text{H}$ σ bond in **A**, **B** adds to the weak $\text{N}=\text{O}$ π bond in HNO ,

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Addition reaction of the $\text{N}-\text{H}$ σ bond in **A**, **B** to the weak $\text{N}=\text{O}$ π bond in HNO

forming an unstable $\text{N}-\text{N}$ single bond, with the hydrogen atom adding to the oxygen atom. Since **C** lacks hydrogen atoms, it cannot undergo this addition reaction, so Statement 8 is incorrect.

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Statement 8 is incorrect

The transformation from **E**₁ to **E**₂ involves a dehydration reaction, clearly a process where the $\text{N}-\text{N}$ single bond undergoes elimination to form a double bond. Double bonds only exhibit cis-trans isomerism, while single bonds can rotate, so **E**₁ has more conformational isomers than **E**₂, making Statement 4 correct.

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Statement 4 is correct

Based on the mechanism described above, **E₁**, **F₁** only contain N – N single bonds, and the N = O π bond disappears after the addition reaction, so Statement 5 is incorrect.

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Statement 5 is incorrect

E₂, **F₂** are both products with eliminated N = N double bonds. It is observed that **E₂** can release a molecule of hydrogen chloride to form nitrogen gas, indicating that the structural formula of **E₂** is H – N = N – Cl, which only exhibits cis-trans isomerism, so Statement 6 is correct.

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Structural formula of **E₂** is H – N = N – Cl

CHECKPOINT

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Statement 6 is correct

In summary, the correct statements are 1, 2, 3, 4, and 6. The answer is C.

CHECKPOINT

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Correct statements are 1, 2, 3, 4, and 6. The answer is C